



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

28-06-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,B,D	A,C,D	A,C,D	A,C	A,B,C	B	C	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.25	4.00	12.00	0.31	0.50	1.52	1.00	6.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C,D	B,C,D	A,B,C,D	A,B,D	A,B,C	D	C	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	80.00	85.00	6.00	7.00	4.00	2.00	4.00	384.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,B	A,C	A,C,D	A,D	B,C	A,C,D	A,C	C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.55 or 3.56	8.00	-11.00	4.00	4.73	8.00	7.00	52.00		



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A,B,C,D)

Sol. The equation of the process BC is

$$TV^2 = \text{constant}$$

$$PV^2 = \text{constant}$$

Molar heat capacity of the gas in the process BC is

$$\therefore C = C_V + \frac{R}{1-x}$$

$$= \frac{3R}{2} - \frac{R}{2} = R$$

The work done by the gas during the process BC is

$$\therefore \Delta W_{BC} = \frac{nR\Delta T}{1-x} = \frac{2R(-900)}{1-3} = 900R$$

$$\Delta Q_{AB} = nC_p\Delta T = 2 \times \frac{5R}{2} (1200 - 300) = 4500R$$

$$\Delta Q_{BC} = nC\Delta T = 2 \times R(300 - 1200) = -1800R$$

$$\Delta Q_{CA} = \Delta W_{CA} = nRT_C \ln \left(\frac{V_0}{8V_0} \right)$$

$$= 2R300(-3\ln 2) = -1800R \times 0.7 = -1260R$$

$$\therefore \Delta W_{\text{cycle}} = \Delta Q_{\text{cycle}} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA}$$

$$= 4500R - 1800R - 1260R = 1440R$$

$\therefore$ Efficiency,

$$\eta = \frac{\Delta W_{\text{cycle}}}{\Delta Q_{\text{supplied}}} \times 100 = \frac{1440R}{4500R} \times 100 = 32\%$$

2. Ans. (A,B,D)

3. Ans. (A,C,D)

Sol. Because all the blocks have same area of top surface this means when man will go from one to other, all blocks will be get drowned by same height.

Let us say when man goes from (1) to (2), one will come out of water by distance x and (2) will go into water by distance x .

$$\therefore 2x \Rightarrow (10 - 5) \text{ cm}$$

$$\therefore h_1' \Rightarrow 7.5 \text{ cm}$$

4. Ans. (A,C,D)

Sol. $PV = nRT$

$$P = \frac{nRT}{V} = \frac{nR(T_0 + kV^3)}{V}$$

$$\frac{dP}{dV} = 0, V = \left(\frac{T_0}{2k} \right)^{1/3}$$

$$P_{\min} = \frac{R(T_0 + T_0/2)}{\left(\frac{T_0}{2k} \right)^{1/3}} = \frac{3}{2} RT_0^{2/3} 2^{1/3} k^{1/3}$$

5. Ans. (A,C)

$$\text{Sol. } ab : P_1 V_1^\gamma = P_2 V_2^\gamma \quad \dots(i)$$

$$cd : P_1 V_1^\gamma = P_2 V_2^\gamma \quad \dots(ii)$$

$$bc : P_1 V_1 = P_2 V_2 \quad \dots(iii)$$

Dividing (ii) by (i), we get

$$\frac{V_2}{V_1} = B^{\frac{2-\gamma}{\gamma-1}} = \sqrt{B} \left[\gamma = \frac{5}{3} \right]$$

work done by the gas

$$W = W_{ab} + W_{bc} + W_{cd}$$

$$= \frac{RT}{2} \ln B + \frac{3}{2} P_1 V_1 (1 - B)$$

6. Ans. (A,B,C)

$$\text{Sol. } \vec{a}_0 = (2g\hat{i} + g\hat{j})$$

$$\vec{F}_B = \rho V(\vec{a}_0 - \vec{g}) = 2\rho Vg(\hat{i} + \hat{j})$$

$$\vec{F}_{\text{net}} = \vec{F}_B + m\vec{g} = 2\rho Vg(\hat{i} + \hat{j}) - \rho_0 Vg\hat{j}$$

$$\vec{a}_{\text{net}} = \frac{2\rho}{\rho_0} g(\hat{i} + \hat{j}) - g\hat{j}$$

$$\vec{a}_{\text{relative}} = \vec{a}_{\text{net}} - \vec{a}_0$$

7. Ans. (B)

Sol. $\frac{dT}{dt} \Rightarrow -\frac{K}{C}(T - T_0)$

$$\int_{25}^{20} \frac{dT}{T - T_0} \Rightarrow -\frac{K}{C} \int_0^t dt$$

$$\therefore t = 5 \text{ min.}$$

8. Ans. (C)

Sol. $\frac{dT}{dt} \Rightarrow \frac{K}{C}(3T - (T - T_0))$

$$\int_{25}^{20} \frac{dT}{2T + T_0} \Rightarrow \int_0^t \frac{K}{C} dt$$

$$t = 67.5 \text{ sec}$$

9. Ans. (A)

Sol. $\frac{dQ}{dt} \Rightarrow -KS \frac{dT}{dx} \Rightarrow P$

$$SK_0(1 + ax) \frac{dT}{dx} \Rightarrow -P$$

$$SK_0 \int_T^{T_B} dT = -P \int_x^{\ell_0} \frac{dx}{1 + ax}$$

$$T \Rightarrow \frac{P}{aSK_0} \ell \ln \left[\frac{1 + a\ell_0}{1 + ax} \right] + T_B$$

10. Ans. (B)

Sol. $\lambda_m T = b$

$$T_B \Rightarrow 2000 \text{ K}$$

Put value of $x = 0$ is previous answer.

$$T_A \Rightarrow 17960 \text{ K.}$$

SECTION-II

1. Ans. 2.25

Sol. In time dt shift in centre of mass system (ball + liquid)

$$dS_{CM} = \frac{m_1 ds_1 + m_2 ds_2}{M} = \frac{(\rho_0 V) v dt - (\rho_\ell V) v dt}{M} \dots (i)$$

$$\text{Momentum of system} = M \frac{dS_{CM}}{dt} \dots (ii)$$

$$= (\rho_0 V) v - (V \rho_\ell) v$$

$$\text{Therefore momentum of liquid} = (V \rho_\ell) v = 2.25 \text{ gm/cm}^3.$$

2. Ans. 4.00

Sol. Since rate of heat flow remains same in both the cases, so

$$\int_R^{2R} \frac{dx}{k2\pi x \ell} = \int_{R/4}^R \frac{dx}{nk(2\pi x) \frac{\ell}{2}}$$

$$\Rightarrow nk = 4k \Rightarrow n = 4$$

3. Ans. 12.00

Sol. According to pascal's law pressure variation will take place throughout the liquid.

$\therefore$ Let us assume right piston shifted by y . Left piston shifted by x '.

$$\therefore \frac{kx}{S} = \frac{2K(L-y)}{3S} \dots (i)$$

$\therefore$ Value of liquid will remain same.

$$\therefore 5x = y(3s) \dots (ii)$$

On solving

$$x = 12 \text{ cm}$$

4. Ans. 0.31

Sol. Apply bernoullis equation between the tap & bottle neck.

$$P + \frac{1}{2} \rho V_1^2 + Pgh_1 = P + \frac{1}{2} \rho V_2^2 + Pgh_2$$

$$\frac{1}{2} \rho [V_2^2 - V_1^2] \Rightarrow \rho g(h_2 - h_1)$$

$$\text{Put } h_2 - h_1 = h.$$

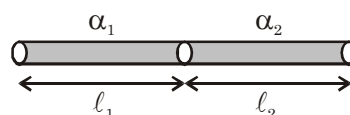
$$V_2 = \frac{L}{A_2}, V_1 = \frac{L}{A_1}$$

where L is volume flow rate

A_1 is the area the tap and A_2 is the area of bottle neck

5. Ans. 0.50

Sol. $\Delta \ell_1 + \Delta \ell_2 = \Delta \ell$



$$\ell_1 \alpha_1 \Delta T + \ell_2 \alpha_2 \Delta T = (\ell_1 + \ell_2) \alpha \Delta T$$

$$\frac{\ell_1}{\ell_1 + \ell_2} = \frac{\alpha - \left(\frac{\ell_2}{\ell_1 + \ell_2} \right) \alpha_2}{\alpha_1} \dots (i)$$

Let $\frac{\ell_1}{\ell_1 + \ell_2} = x$, so $\frac{\ell_2}{\ell_1 + \ell_2} = 1 - x$ (ii)

$$x = \frac{\alpha - (1-x)\alpha_2}{\alpha_1}$$

$$\alpha_1 x = \alpha - \alpha_2 + \alpha_2 x$$

$$x = \left(\frac{\alpha - \alpha_2}{\alpha_1 - \alpha_2} \right) = \frac{17 \times 10^{-6} - 11 \times 10^6}{23 \times 10^6 - 11 \times 10^6}$$

$$= \frac{6}{12} = 0.50$$

6. **Ans. 1.52**

Sol. The centre of mass of the rod is at distance

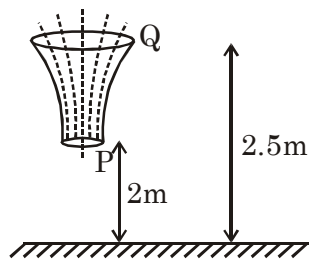
$$\frac{(y-x)}{2} \text{ from the brim.}$$

$$\left(\rho_{al} g V - \rho_w g \frac{V}{2} \right) x = mg \frac{(y-x)}{2}$$

$$\Rightarrow 4/x = 1.52$$

7. **Ans. 1.00**

Sol. $P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g (h_2 - h_1)$



$$= \frac{1}{2} \times 10^3 (1^2 - 3^2) + 10^3 \times 10(0.5)$$

$$= -4 \times 10^3 + 5 \times 10^3 = 1000 \text{ J/m}^2$$

8. **Ans. 6.00**

Sol. $x_{cm} = \frac{\int dm \cdot x}{\int dm} = \frac{2L}{3}$

Mass of rod;

$$M = \int_0^L \lambda_0 x dx = \frac{10^3}{2}$$

Torque about O;

$$F_B \frac{L}{2} \cos 37^\circ - Mg \frac{2L}{3} \cos 37^\circ = TL \cos 37^\circ$$

$$10^3 \pi \left(\frac{1}{\pi} \right) \times 10 \frac{1}{2} - \frac{10^4}{2} \times \frac{2}{3} = T$$

$$T = \frac{10^4}{6} \text{ N}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans.(C)**
2. **Ans.(C,D)**
3. **Ans.(B,C,D)**
4. **Ans.(A,B,C,D)**
5. **Ans.(A,B,D)**
6. **Ans.(A,B,C)**
7. **Ans.(D)**
8. **Ans.(C)**
9. **Ans.(B)**
10. **Ans.(C)**

SECTION-II

1. **Ans.(80.00)**
2. **Ans.(85.00)**
3. **Ans.(6.00)**
4. **Ans.(7.00)**
5. **Ans.(4.00)**
6. **Ans.(2.00)**
7. **Ans.(4.00)**
8. **Ans. (384.00)**

Sol. $12 \times 8 \times 4 = 384$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,D)

Sol. Points are $(0, 0)$ $(t^2, 2t)$ & $\left(\frac{1}{t^2}, -\frac{2}{t}\right)$

$$h = \frac{t^2 + \frac{1}{t^2}}{3}$$

$$\Rightarrow \left(t - \frac{1}{t}\right)^2 + 2 = 3h$$

$$K = \frac{2\left(t - \frac{1}{t}\right)}{3}$$

$$\Rightarrow t - \frac{1}{t} = \frac{3K}{2}$$

$$\frac{9y^2}{4} + 2 = 3h$$

$$9y^2 = 12h - 8$$

$$y^2 = \frac{12}{9}\left(h - \frac{2}{3}\right)$$

$$y^2 = \frac{4}{3}\left(h - \frac{2}{3}\right)$$

2. Ans. (A,B)

Sol. $\therefore \frac{x^2 + \beta x + \beta^2}{3} \leq x^2 - \alpha x + \alpha^2 \leq 3(x^2 + \beta x + \beta^2)$

$$\underbrace{\quad}_{(i)} \quad \underbrace{\quad}_{(ii)}$$

$$(\because x^2 + \beta x + \beta^2 > 0 \forall x \in \mathbb{R})$$

$$(i) \Rightarrow 3x^2 - 3\alpha x + 3\alpha^2 - x^2 - \beta x + \beta^2 \geq 0$$

$$\Rightarrow 2x^2 - (3\alpha + \beta)x + (3\alpha^2 - \beta^2) \geq 0$$

$$\Rightarrow (3\alpha + \beta)^2 - 8(3\alpha^2 - \beta^2) \leq 0$$

$$\Rightarrow -15\alpha^2 + 9\beta^2 + 6\alpha\beta \leq 0 \quad \text{div. by } 3\beta^2$$

$$\Rightarrow -5\left(\frac{\alpha}{\beta}\right)^2 + 2\frac{\alpha}{\beta} + 3 \leq 0$$

$$\Rightarrow \left(5\frac{\alpha}{\beta} + 3\right)\left(\frac{\alpha}{\beta} - 1\right) \geq 0$$

$$\Rightarrow \frac{\alpha}{\beta} \leq -\frac{3}{5} \text{ or } \frac{\alpha}{\beta} \geq 1 \quad \dots(1)$$

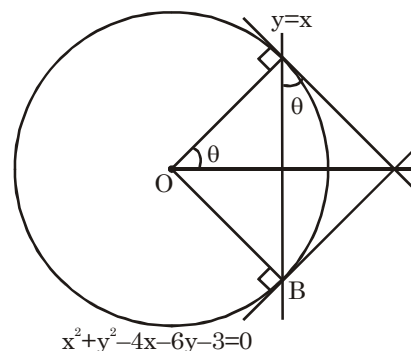
$$\text{similarly (ii)} \Rightarrow \left(\frac{\alpha}{\beta} - 1\right)\left(\frac{3\alpha}{\beta} + 5\right) \leq 0$$

$$\Rightarrow -\frac{5}{3} \leq \frac{\alpha}{\beta} \leq 1 \quad \dots(2)$$

$$\therefore \text{from (1) and (2)} \quad -\frac{5}{3} \leq \frac{\alpha}{\beta} \leq -\frac{3}{5} \text{ or } \frac{\alpha}{\beta} = 1$$

3. Ans. (A,C)

Sol.



4. Ans. (A,C,D)

Sol. Equation of tangent to $y^2 = 4x$ at $(4, 4)$ is $x - 2y + 4 = 0$

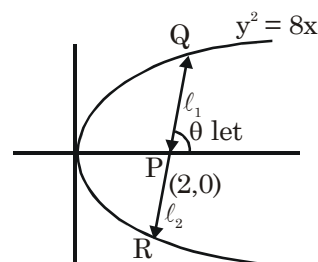
Now, distance of the center of the variable circle from this tangent is equal to its distance from $(1, 0)$. Hence, locus of the center will be a parabola having focus at $(1, 0)$ & $x - 2y + 4 = 0$ as the directrix.

So, length of latus rectum

$$= 2 \times \frac{5}{\sqrt{5}} = 2\sqrt{5}$$

5. Ans. (A,D)

Sol.



Let equation of QR be

$$\frac{x-2}{\cos \theta} = \frac{y}{\sin \theta} = \ell$$

$$x = \ell \cos \theta + 2$$

$$y = \ell \sin \theta$$

$$\text{if } (\ell \cos \theta + 2, \ell \sin \theta) \text{ lies on } y^2 = 8x$$

$$\Rightarrow (\ell \sin \theta)^2 = 8(\ell \cos \theta + 2)$$

$$\Rightarrow \ell^2 \sin^2 \theta - 8 \cos \theta \cdot \ell - 16 = 0$$

here roots are ℓ_1 and $-\ell_2$

$$\Rightarrow \ell_1 - \ell_2 = \frac{8 \cos \theta}{\sin^2 \theta} \text{ and } -\ell_1 \ell_2 = -\frac{16}{\sin^2 \theta}$$

$$\text{Now harmonic mean} = \frac{2\ell_1 \ell_2}{\ell_1 + \ell_2}$$

$$= \frac{2\ell_1 \ell_2}{\sqrt{(\ell_1 - \ell_2)^2 + 4\ell_1 \ell_2}}$$

$$= 4$$

$$\text{we know that } G^2 = AH \Rightarrow G^2 = 4A$$

$$\Rightarrow G \propto \sqrt{A}$$

6. **Ans. (B,C)**

$$R \cos A = r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\cos A = \cos B + \cos C - 1 + \cos A$$

Paragraph for Question 7 & 8

Sol. For pair of straight lines,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$(2)(2)(a) + 2(2)\left(\frac{5}{2}\right)\left(\frac{5}{2}\right) - 2\left(\frac{5}{2}\right)^2 - 2(2)^2 - a\left(\frac{5}{2}\right)^2 = 0$$

$$4a + 25 - \frac{25}{2} - 8 - \frac{25a}{4} = 0$$

$$\frac{9a}{4} = \frac{9}{2} \Rightarrow a = 2$$

$$2x^2 + 5xy + 2y^2 \quad 4x + 5y + 2 = 0$$

$$2x + y + 2 = 0 \text{ or } x + 2y + 1 = 0$$

$$\text{Line } bx + y + 5 = 0$$

7. **Ans. (A,C,D)**

Sol. Form a triangle then we have four circles touching all sides In circle & three excircles

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \\ b & 1 & 5 \end{vmatrix} \neq 0$$

$$2(9) - (5 - b) + 2(1 - 2b) \neq 0$$

$$15 \neq 3b \Rightarrow b \neq 5$$

8. **Ans. (A,C)**

Sol. Possible when the lines are parallel means

$$-2 = -b \text{ or } -b = -\frac{1}{2}$$

$$b = 2 \text{ or } b = \frac{1}{2}$$

Paragraph for Question 9 & 10

9. **Ans. (C,D)**

Sol. Using $A > G$ pairwise

$$\frac{(a_1 + a_2 - a_3) + (a_2 + a_3 - a_1)}{2} > \sqrt{(a_1 + a_2 - a_3)(a_2 + a_3 - a_1)}$$

$$\Rightarrow a_2 > ((a_1 + a_2 - a_3)(a_2 + a_3 - a_1))^{1/2}$$

Similarly

$$a_3 > ((a_2 + a_3 - a_1)(a_3 + a_1 - a_2))^{1/2}$$

$$\& a_1 > ((a_1 + a_2 - a_3)(a_3 + a_1 - a_2))^{1/2}$$

Multiplying above result

10. **Ans. (A,B,C)**

Sol. $\therefore$ Use $A > H$

$$\therefore \frac{a_1 + a_2 + \dots + a_n}{n} > \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}$$

$$\Rightarrow S > \frac{n^2}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}} \therefore (A) \text{ is +ve}$$

$$(B) \therefore a_1 + a_2 + \dots + a_n$$

$$= \frac{n}{2}(a_1 + a_n) = n \left(\frac{a_1 + a_n}{2} \right) > n \cdot \frac{2a_1 a_n}{a_1 + a_n} (A > H)$$

$$(C) a_1 + a_2 + \dots + a_n$$

$$= \frac{n}{2}[2a_1 + (n-1)d] = n \left[\frac{2a_1 + (n-1)d}{2} \right] > n \cdot \sqrt{2a_1(n-1)d}$$

$$(A > G)$$

$$(D) \text{ From option (B)}$$

$$S = n \left(\frac{a_1 + a_n}{2} \right) > n \sqrt{a_1 a_n} (A > G)$$

SECTION-II

1. **Ans. 3.55 or 3.56**

Sol. Intersection point of tangent at vertex and tangent is $(-1, 1)$

Now equation of line which is $\perp$ from given tangent and passing through $(-1, 1)$ is

$$y - 1 = -\frac{1}{2}(x + 1) \Rightarrow 2y - 2 = -x - 1$$

$$x + 2y = 1$$

this line $x + 2y = 1$ and equation of axis satisfied the focus

$$\therefore x + 2y = 1 \text{ and } x = y \Rightarrow x = \frac{1}{3}, y = \frac{1}{3}$$

Now distance between focus and tangent at vertex is

$$a = \left| \frac{\frac{1}{3} + \frac{1}{3}}{\sqrt{2}} \right| \Rightarrow 4a = \frac{8}{3\sqrt{2}}$$

$$= (4a)^2 = \frac{64}{18} \Rightarrow L = \frac{32}{9}$$

2. Ans. 8.00

Sol. $x + y \geq 2\sqrt{xy}$

$$y + z \geq 2\sqrt{yz}$$

$$z + x \geq 2\sqrt{zx}$$

multiplying above

$$(x + y)(y + z)(z + x) \geq 8xyz$$

$$(1 - z)(1 - x)(1 - y) \geq 8xyz$$

$$\Rightarrow \frac{(1-x)(1-y)(1-z)}{xyz} \geq 8$$

3. Ans. -11.00

Sol. Let $\Delta(x) = \begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} = Ax^4 + Bx^3 + Cx^2 + Dx + E$

then by RHS

$$\Delta(1) + \Delta'(1) = 5A + 4B + 3C + 2D + E$$

Now, LHS

$$\Delta(1) = 0$$

$$\Delta'(x) = \begin{vmatrix} 1 & 0 & 1 \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} + \begin{vmatrix} x & 2 & x \\ 2x & 1 & 0 \\ x & x & 6 \end{vmatrix} + \begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ 1 & 1 & 0 \end{vmatrix}$$

$$\Delta'(1) = -11$$

$$\therefore 5A + 4B + 3C + 2D + E = -11$$

4. Ans. 4.00

Sol.
$$\frac{\cos 6^\circ}{(\cos 14^\circ + 2\cos^2 7^\circ)\cos 83^\circ \cos 21^\circ \cos 42^\circ}$$

$$= \frac{\cos 6^\circ}{(1 - 2\sin^2 7^\circ + 2(1 - \sin^2 7^\circ))\sin 7^\circ \cos 21^\circ \cos 42^\circ}$$

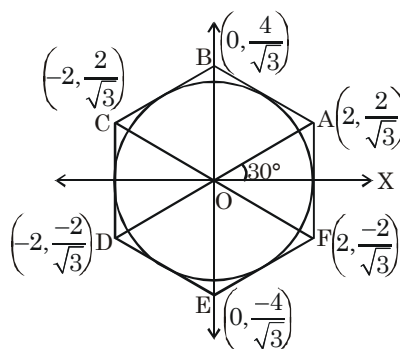
$$= \frac{\cos 6^\circ}{(3 - 4\sin^2 7^\circ)\sin 7^\circ \cos 21^\circ \cos 42^\circ}$$

$$= \frac{2\cos 6^\circ}{2\sin 21^\circ \cos 21^\circ \cos 42^\circ}$$

$$= \frac{4\cos 6^\circ}{2\sin 42^\circ \cos 42^\circ} = \frac{4\cos 6^\circ}{\sin 84^\circ} = 4$$

5. Ans. 4.73

Sol.



Shifting origin to $\left(1, \frac{1}{\sqrt{3}}\right)$ and $A(3, \sqrt{3})$

becomes $A\left(2, \frac{2}{\sqrt{3}}\right)$. F is image of A about x-axis. C is image of F about origin. D is image of A about origin

$\Rightarrow$ actual vertices will be $A(3, \sqrt{3})$,

$$B\left(1, \frac{5}{\sqrt{3}}\right), C(-1, \sqrt{3}), D\left(-1, -\frac{1}{\sqrt{3}}\right),$$

$$E(1, -\sqrt{3}), F\left(3, -\frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow \sum (x_i + y_i) = 3 + \sqrt{3}$$

6. Ans. 8.00

Sol. $C_1 : x^2 + y^2 = 32$

$$C_2 : x^2 + y^2 - 4x - 4y - 10 = 0$$

$\therefore$ required P and Q lie on circle

$$x^2 + y^2 - 64 + \lambda(x + y - 9) = 0$$

$$x^2 + y^2 + \lambda x + \lambda y - 9\lambda - 64 = 0$$

for smallest circle centre $\left(-\frac{\lambda}{2}, -\frac{\lambda}{2}\right)$ lie on $x + y - 9 = 0$

$$\text{required circle } x^2 + y^2 - 9x - 9y + 17 = 0$$

7. Ans. 7.00

Sol. $x^2y^2 + x^2 + y^2 + 1 + 9 - 6x - 6y = 0$

$$\Rightarrow x^2y^2 - 2xy + 1 + x^2 + y^2 + 9 + 2xy - 6x - 6y = 0$$

$$\Rightarrow (xy - 1)^2 + (x + y - 3)^2 = 0$$

$$\Rightarrow xy = 1 \text{ and } x + y = 3$$

$$\Rightarrow x^2 + y^2 = (x + y)^2 - 2xy = 7$$

8. Ans. 52.00

Sol. Both circles intersect at two distinct points

$$\text{use } |r_1 - r_2| < c_1c_2 < r_1 + r_2$$

$\therefore c$ is positive integer

$$\therefore c \in \{1, 2, 3, \dots, 52\}$$